

Labwork 10.2 - Magic of Recursive World

General Information and Submission Rules:

- By the end of the labwork, you must submit it to Yulearn as **studentID_fullname_labwork8.c**.
- Incorrect naming or file format will result in a loss of 5 points from your total grade.
- Only use concepts covered in the labs so far.

Question 1 - Display Message (10p)

- Create a new function (**welcome**) that displays a welcome message and a list of all available operations to the user.
- This function will expect a decision from a user so that it will execute related functions based on this choice until the user exits.
- Make sure to print each line separately.

```
Welcome to Labwork 10 Magic of Recursive World! By using this machine you can do:
1. The Missing Compass
2. The Weaver's Pattern
3. Separate Characters with Magic
4. Exit
Make your choice:
```

Question 2 - The Missing Compass (35p)

- A group of explorers is lost in a vast desert. They have a map with several marked locations, each at a different distance (the array values).
- Their GPS has broken, but they know their current coordinate (the query integer). They need to find the location that is closest to their current location to survive the heat.
- Create a new recursive function (**missing_compass**) that takes an array of numbers (that represents distances) and a query integer.

- In addition to this, you need a random number generator function which is given below for creating distances.
- In the end, print the closest location.

```
#include <time.h>
int generateRandom(int limit) {
    srand(time(NULL));
    return rand() % (limit + 1);
}
```

Generated Array: [123, 456, 789, 1000, 300, 800]

Output Example:

>> Query: 200

>> Closest location is 123

Question 3 - The Weaver's Pattern (30p)

- A royal weaver is tasked with creating a "Fractal String". For a given intensity N , the weaver must print a sequence where N is in the middle with adjacent $N-1$.
- Ex. for $N=3$, the pattern is
1 2 1 3 1 2 1
For $N=4$, the pattern is
1 2 1 3 1 4 1 3 1 2 1
- Create a recursive function (**weaver_pattern**) that takes the number N that shows the intensity of the pattern.
- In the end, print the designed pattern.

```
>> Please enter N: 3
1 2 1 3 1 2 1
```

Question 4 - Separate Characters with Magic (25p)

- A tiny fairy named Elara has a very specific kind of magic. Elara is the Fairy of Connections. She believes that every letter in a word is too lonely if it's side by side with another letter.
- To help them feel special, she uses her wand to place a Magic Star (*) between every single character.
- Create a recursive function (**star_magic**) that takes the string. (The string can be created in main directly)

- In the end, print the word after a Magic Star.

```
>> h*e*l*l*o
```